BP-ADMM Mathematical Notes (ECG Denoising)

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Toy Example – Input & Processing

Objective

- Understand how the proposed method cleans and reconstructs an ECG signal using a simple example.

Input (Clean ECG)

$$\boldsymbol{x} = [0,\ 1,\ 3,\ 7,\ 3,\ 1,\ 0,\ 0]$$

- Represents a small ECG segment
- Peak corresponds to QRS complex

Noise Addition (Realistic Scenario)

- Baseline wander (low frequency):

$$\boldsymbol{f} = [1,\ 1,\ 1,\ 1,\ 1,\ 1,\ 1,\ 1]$$

- Gaussian noise added:

$$\boldsymbol{y} = \boldsymbol{x} + \boldsymbol{f} + \boldsymbol{n}$$

Baseline Removal

- Low-pass filter estimates baseline: $\hat{\boldsymbol{f}}$
- Subtract baseline:

$$\boldsymbol{y}_c = \boldsymbol{y} - \hat{\boldsymbol{f}}$$

Result: ✓ Removes drift ✓ Preserves ECG shape

Key Idea

$$\boldsymbol{x} = \boldsymbol{\Psi}\boldsymbol{s}$$

Compression, Reconstruction & Output

Compressed Sensing

- Instead of storing all samples, take fewer measurements:

$$\boldsymbol{y}_m = \boldsymbol{M}\boldsymbol{x}$$

- Data size reduced (8 $\rightarrow$ 4 samples)

Reconstruction Model

- Recover ECG by solving:

$$\min_{\boldsymbol{x}} \frac{1}{2} \|\boldsymbol{y}_m - \boldsymbol{M}\boldsymbol{x}\|_2^2 + \lambda \|\boldsymbol{x}\|_1$$

- First term $\rightarrow$ fits measurements
- Second term $\rightarrow$ enforces sparsity and removes noise

Output (Recovered ECG)

$$\hat{\boldsymbol{x}} \approx [0,\ 1,\ 3,\ 7,\ 3,\ 1,\ 0,\ 0]$$

Result: ✓ Baseline removed ✓ Noise suppressed ✓ ECG morphology preserved

1) ECG observation model

Equation

$$\boldsymbol{y} = \boldsymbol{x} + \boldsymbol{f} + \boldsymbol{n}$$

- $\boldsymbol{y}$: observed ECG (what we measure)
- $\boldsymbol{x}$: clean ECG (what we want)
- $\boldsymbol{f}$: baseline wander (slow low-frequency drift)
- $\boldsymbol{n}$: random noise (Gaussian white noise)

Meaning: the measured signal is a sum of the true ECG + baseline drift + noise.

2) Compressed sensing measurement model

Equation

$$\boldsymbol{y} = \boldsymbol{M}\boldsymbol{x} + \boldsymbol{f} + \boldsymbol{n}$$

- $\boldsymbol{M} \in \mathbb{R}^{m \times n}$, with $m \ll n$
- $\boldsymbol{M}\boldsymbol{x}$ means we measure the ECG through a sensing/measurement matrix $\boldsymbol{M}$
- (fewer samples than the signal length)

Why this change from $\boldsymbol{x}$ to $\boldsymbol{M}\boldsymbol{x}$?

Because in compressed sensing, we don't observe the full signal directly; we observe a compressed version of it.

3) Sparse representation of ECG

Key assumption

ECG becomes sparse in some transform domain:

$$\boldsymbol{x} = \boldsymbol{\Psi}\boldsymbol{s}$$

- $\boldsymbol{\Psi}$: dictionary / basis (e.g., STFT basis)
- $\boldsymbol{s}$: sparse coefficients (most entries are 0 or near 0)

Substitute into measurement

Starting from the compressed model (ignoring $\boldsymbol{f}, \boldsymbol{n}$ for this step):

$$\boldsymbol{y} \approx \boldsymbol{M}\boldsymbol{x} = \boldsymbol{M}(\boldsymbol{\Psi}\boldsymbol{s}) = (\boldsymbol{M}\boldsymbol{\Psi})\boldsymbol{s}$$

So effectively, we solve for sparse $\boldsymbol{s}$, then reconstruct $\boldsymbol{x} = \boldsymbol{\Psi}\boldsymbol{s}$.

4) Sparse recovery as optimization (constrained)

Basis Pursuit Denoising form

$$\min_{\boldsymbol{x}} \|\boldsymbol{x}\|_1 \quad \text{s.t.} \quad \|\boldsymbol{y} - \boldsymbol{M}\boldsymbol{x}\|_2^2 \leq \epsilon$$

- $\|\boldsymbol{x}\|_1 = \sum_i |x_i|$: encourages sparsity
- $\|\boldsymbol{y} - \boldsymbol{M}\boldsymbol{x}\|_2^2$: data fidelity (fit measurements)
- ϵ: allowed error (because noise exists)

Why L_1? Because L_1 promotes sparse solutions but stays convex (solvable reliably).

5) Unconstrained Lagrangian form

The constrained problem is often converted into an unconstrained "penalty" form:

$$\min_{\boldsymbol{x}} \frac{1}{2}\|\boldsymbol{y} - \boldsymbol{M}\boldsymbol{x}\|_2^2 + \lambda\|\boldsymbol{x}\|_1$$

- $\lambda > 0$ trades off:
  - small $\|\boldsymbol{y} - \boldsymbol{M}\boldsymbol{x}\|_2^2$ (fit)
  - sparse $\boldsymbol{x}$ (via $\|\boldsymbol{x}\|_1$)

How to think of it:
Instead of "must be within ϵ", we say "pay a cost if error is big" + "pay a cost if not sparse".

6) Baseline wander estimation

Baseline wander is mostly low frequency, so we use a low-pass filter:

$$\hat{\boldsymbol{f}} = \boldsymbol{H}_{LP}(\boldsymbol{y})$$

Then baseline corrected:

$$\boldsymbol{y}_c = \boldsymbol{y} - \hat{\boldsymbol{f}}$$

So after correction, the main remaining corruption is Gaussian noise.

7) Combined objective (as shown)

$$\min_{\boldsymbol{x}} \frac{1}{2}\|\boldsymbol{y} - \boldsymbol{M}\boldsymbol{x}\|_2^2 + \lambda\|\boldsymbol{H}\boldsymbol{x} - \boldsymbol{y}\|_1$$

- first term: reconstruction error
- second term: an L_1 penalty after applying operator/filter $\boldsymbol{H}$

Intuition: L_1 on a residual is robust to outliers and connects to sparse modeling.

8) Variable splitting to use ADMM

ADMM works best when the objective is separable. Introduce an auxiliary variable:

$$\boldsymbol{z} = \boldsymbol{A}\boldsymbol{x}$$

Then rewrite:

$$\min_{\boldsymbol{x},\boldsymbol{z}} \frac{1}{2}\|\boldsymbol{y} - \boldsymbol{M}\boldsymbol{x}\|_2^2 + \lambda\|\boldsymbol{z}\|_1 \quad \text{s.t.} \quad \boldsymbol{z} = \boldsymbol{A}\boldsymbol{x}$$

Now:

- quadratic term depends on $\boldsymbol{x}$
- L_1 term depends on $\boldsymbol{z}$
- constraint links them

9) Augmented Lagrangian (core ADMM idea)

To enforce constraints, we build an augmented Lagrangian:

$$\mathcal{L}(\boldsymbol{x}, \boldsymbol{z}, \boldsymbol{\lambda}, \boldsymbol{v}) = \|\boldsymbol{z}\|_1 + \boldsymbol{\lambda}^T(\boldsymbol{M}\boldsymbol{x} - \boldsymbol{y}) + \frac{\rho}{2}\|\boldsymbol{M}\boldsymbol{x} - \boldsymbol{y}\|_2^2 + \boldsymbol{v}^T(\boldsymbol{A}\boldsymbol{x} - \boldsymbol{z}) + \frac{\mu}{2}\|\boldsymbol{A}\boldsymbol{x} - \boldsymbol{z}\|_2^2$$

- $\boldsymbol{\lambda}, \boldsymbol{v}$: dual variables (multipliers)
- ρ, μ: penalty parameters

Why add squared penalty terms?

They make constraint violations expensive and stabilize convergence.

10) ADMM updates (the iterative equations)

ADMM repeats 3 main steps:

$$\boldsymbol{x}\text{-update} \;\rightarrow\; \boldsymbol{z}\text{-update} \;\rightarrow\; \text{dual updates}$$

Step 1: x-update (solve a quadratic)

Fix $\boldsymbol{z}$, minimize w.r.t. $\boldsymbol{x}$:

$$\min_{\boldsymbol{x}} \frac{\rho}{2}\|\boldsymbol{M}\boldsymbol{x} - \boldsymbol{y}\|_2^2 + \frac{\mu}{2}\|\boldsymbol{A}\boldsymbol{x} - \boldsymbol{z}\|_2^2 + \boldsymbol{\lambda}^T(\boldsymbol{M}\boldsymbol{x} - \boldsymbol{y}) + \boldsymbol{v}^T(\boldsymbol{A}\boldsymbol{x} - \boldsymbol{z})$$

Take derivative and set to zero:

$$\rho\, \boldsymbol{M}^T(\boldsymbol{M}\boldsymbol{x} - \boldsymbol{y}) + \mu\, \boldsymbol{A}^T(\boldsymbol{A}\boldsymbol{x} - \boldsymbol{z}) + \boldsymbol{M}^T\boldsymbol{\lambda} + \boldsymbol{A}^T\boldsymbol{v} = 0$$

Step 1 continued: expand and solve for $\boldsymbol{x}$

Expand:

$$\rho\,\boldsymbol{M}^T(\boldsymbol{M}\boldsymbol{x}-\boldsymbol{y}) = \rho(\boldsymbol{M}^T\boldsymbol{M}\boldsymbol{x}-\boldsymbol{M}^T\boldsymbol{y}), \quad \mu\,\boldsymbol{A}^T(\boldsymbol{A}\boldsymbol{x}-\boldsymbol{z}) = \mu(\boldsymbol{A}^T\boldsymbol{A}\boldsymbol{x}-\boldsymbol{A}^T\boldsymbol{z})$$

So:

$$\rho\,\boldsymbol{M}^T\boldsymbol{M}\boldsymbol{x} - \rho\,\boldsymbol{M}^T\boldsymbol{y} + \mu\,\boldsymbol{A}^T\boldsymbol{A}\boldsymbol{x} - \mu\,\boldsymbol{A}^T\boldsymbol{z} + \boldsymbol{M}^T\boldsymbol{\lambda} + \boldsymbol{A}^T\boldsymbol{v} = 0$$

Move non-$\boldsymbol{x}$ terms:

$$(\rho\,\boldsymbol{M}^T\boldsymbol{M} + \mu\,\boldsymbol{A}^T\boldsymbol{A})\boldsymbol{x} = \mu\,\boldsymbol{A}^T\boldsymbol{z} + \rho\,\boldsymbol{M}^T\boldsymbol{y} - \boldsymbol{M}^T\boldsymbol{\lambda} - \boldsymbol{A}^T\boldsymbol{v}$$

Finally:

$$\boldsymbol{x}^{k+1} = (\rho\,\boldsymbol{M}^T\boldsymbol{M} + \mu\,\boldsymbol{A}^T\boldsymbol{A})^{-1}\left(\mu\,\boldsymbol{A}^T\boldsymbol{z}^k + \rho\,\boldsymbol{M}^T\boldsymbol{y} - \boldsymbol{M}^T\boldsymbol{\lambda}^k - \boldsymbol{A}^T\boldsymbol{v}^k\right)$$

Step 2: z-update (soft thresholding)

Fix $\boldsymbol{x}$, minimize w.r.t. $\boldsymbol{z}$:

$$\min_{\boldsymbol{z}} \|\boldsymbol{z}\|_1 + \frac{\rho}{2}\left\|\boldsymbol{A}\boldsymbol{x}^{k+1} - \boldsymbol{z} - \frac{\boldsymbol{\lambda}}{\rho}\right\|_2^2$$

This is the classic "L_1 + quadratic" proximal problem $\Rightarrow$ solved by soft-thresholding.

Soft-threshold operator:

$$S(x, T) = \begin{cases} x - T, & x > T \\ 0, & |x| \leq T \\ x + T, & x < -T \end{cases}$$

Then:

$$\boldsymbol{z}^{k+1} = S_{1/\rho}\left(\boldsymbol{A}\boldsymbol{x}^{k+1} + \frac{\boldsymbol{\lambda}}{\rho}\right)$$

Step 3: Dual variable updates

Dual updates "accumulate" constraint violations:

$$\boldsymbol{\lambda}^{k+1} = \boldsymbol{\lambda}^{k} + \rho(\boldsymbol{M}\boldsymbol{x}^{k} - \boldsymbol{y})$$

$$\boldsymbol{v}^{k+1} = \boldsymbol{v}^{k} + \mu(\boldsymbol{A}\boldsymbol{x}^{k} - \boldsymbol{z}^{k})$$

If constraints aren't satisfied, multipliers adjust to force them closer in the next iteration.

11) Convergence criteria (when to stop)

Stop when:

- **Primal residual** is small (constraints satisfied):

$$\boldsymbol{Ax} - \boldsymbol{z} \approx \mathbf{0}, \qquad \boldsymbol{Mx} - \boldsymbol{y} \approx \mathbf{0}$$

- **Dual residual** is small (updates become tiny)

12) Performance metrics

SNR

$$\mathrm{SNR} = 10 \log_{10} \left(\frac{\sum x(i)^2}{\sum \left(x(i) - \hat{x}(i)\right)^2} \right)$$

MSE

$$\mathrm{MSE} = \frac{1}{N} \sum \left(x(i) - \hat{x}(i)\right)^2$$

Both compare clean $\boldsymbol{x}$ vs reconstructed $\hat{\boldsymbol{x}}$.

One-line summary of the whole math pipeline

$\boldsymbol{y} \to$ sparse model $\to$ BPDN objective $\to$ ADMM split ($\boldsymbol{z} = \boldsymbol{A}\boldsymbol{x}$) $\to$ $\boldsymbol{x}$-update $\to$ $\boldsymbol{z}$-update (soft-t